# Final Project: Pick a Project

Computer Science - Week 10

Oct 3, 2026 - Version 0.0.2

Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

| Role | Name | Email |
|---|---|---|
| **Developer (You)** | Zara Martinez | zaram@udel.edu |
| **Friend (optional)** | | |

## Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

Aqueous Reaction Simulator. This can work in two different ways: precipitate* and single displacement. This program will mainly use an activity series that assigns known metals to numbers then will use an equality operator. The program will then provide an explanation on why it does or does not work.

*I don't really know how I'll do this yet, but I'll probably make a seperate database for soluble and non-soluble chemicals.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

- *Google*

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

*It will be similar to another site I use frequently, called the Integral Calculator. It will be very general, since it'll utilize general chemistry topics. There will be two options, either to do*

*single displacement or precipitate solutions. Users will have the option to input their reactants (2 required). It will give the expected products and say if it will or will not work.*

# Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

Input reactants and output expected reactants.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

Reactants 2x

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

Compare the input against a database/list

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

Expected products

# Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Reactants, Products, Compounds

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

*Reactant molar mass, Expected products molar mass, How reactants will react physically/chemically*

# Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

*Making the actual drafter part, unfortunately.*

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

*I'm going to attempt to not use AI as much as possible, only if I need help on the project. About 30% of the project will use AI.*

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

*Learn Drafter deeply and impress the graders with a creative UI.*

# Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

| |
|---|
| I want to build a simple web tool to help students learn about aqueous solutions using a simulator where the students input the reactants to get the products. What are three things this tool should have? |

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

| |
|---|
| - *Provide dropdown menus for set reactants and products*<br>- *Visual Animations*<br>- *Practice Mode* |

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

| |
|---|
| *Create a practice mode and show a balanced equation, then ask if a reaction will occur. Provide an explanation of why it does.* |

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

| |
|---|
| *Not a lot has changed, but if I have extra time I can definitely add more concepts like acids/bases.* |

# Part 6: Resource Planning

1. What resources will you use if you get stuck?

*I will heavily rely on the Drafter website in case I get stuck (that's what I did for Project 2). If I get REALLY stuck, I'll ask ChatGPT for help.*

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

*I'll look to see what ChatGPT wrote to see if I can fix myself. Otherwise, I'll ask ChatGPT what went wrong.*

# Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I want to build a simple web tool to help students learn about aqueous solutions using a simulator where the students input the reactants to get the products. What are three things this tool should have?

For a simple web tool that helps students learn about aqueous solutions by entering reactants and predicting products, I'd recommend these three essential features:

## 1. Interactive Reaction Simulator 🧪

edu.rsc.org

Students enter two reactants, such as aqueous lead(II) nitrate and potassium iodide, and the simulator generates the products.

- Provide dropdown menus for selecting reactants and their states (aq, s, l, or g).
- Display the predicted products and identify whether a precipitate, gas, or water forms.
- Use visual animations, such as a solid appearing when a precipitate forms.

Learning goal: Help students recognize and predict different types of reactions in aqueous solutions.

## 2. Step-by-Step Explanations and Equation Balancing

After students submit their reactants, show how the reaction works rather than just revealing the answer.

- Display the balanced molecular equation.
- Break it down into a complete ionic equation and a net ionic equation.
- Explain which ions are spectators and why a precipitate forms, using solubility rules.

For example:

Pb(NO3)2(aq)+2KI(aq)\mathrm{Pb(NO_3)_2(aq)+2KI(aq)}
→PbI2(s)+2KNO3(aq)\rightarrow \mathrm{PbI_2(s)+2KNO_3(aq)}

The tool would explain that lead(II) iodide forms a yellow precipitate, while potassium and nitrate ions remain dissolved.

Learning goal: Help students understand the chemistry behind the reaction, not simply memorize products.

## 3. Practice Mode with Instant Feedback

Give students opportunities to test their understanding independently.

- Generate random reaction problems with different reactants.
- Let students predict the products before revealing the solution.
- Provide instant feedback, hints, and explanations for incorrect answers.
- Track scores or progress so students can see how they improve.

Learning goal: Reinforce learning through practice and help students identify concepts they need to review.

Prototype demonstration: a complete tool should validate all reactant combinations, balance equations automatically, and account for reaction conditions.

My recommendation: Build the reaction simulator first, then add explanations and practice mode. This gives students an engaging way to explore reactions while ensuring that the tool teaches the reasoning behind their predictions.

I think the visual animations is a bit much for an introductory computer project

I agree! For an introductory computer project, you can keep it much simpler while still making it useful for chemistry students. You don't need animations or complicated visual effects.

Here are three features I'd recommend instead:

## 1. Reactant Input and Product Prediction

- Let students select two aqueous reactants from dropdown menus.
- Display the predicted chemical products when they click a button.
- Show the correct chemical formulas and physical states, such as (aq) or (s).

## 2. Balanced Equations and Simple Explanations

- Display the balanced chemical equation for each reaction.
- Identify whether a precipitate forms or whether no reaction occurs.
- Include a short explanation based on solubility rules.

## 3. Practice Quiz with Feedback

- Give students practice problems with different reactant combinations.
- Let them submit their predictions before revealing the answer.
- Tell them whether their answer is correct and show the correct solution if they get it wrong.

These features are achievable with basic HTML, CSS, and JavaScript. You could store a small set of known reactions in a JavaScript object and use conditional logic to display the corresponding products and explanations.

The key is to focus on helping students predict reactions, not on making the website visually complicated.

# Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again.

Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.